Phoenix Nuclear Power Plant (PNPP) Station Documentation & Reference Dossier

"From the desert, power without end." — Desert Horizons Energy station motto, cast in bronze at the Unit 1 turbine deck entrance since 1999.

Operator: Desert Horizons Energy Corporation (DHE)
Location: Sonoran Desert, southwestern Maricopa County, Arizona, USA
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NOTE 1: This is a fictional document. Desert Horizons Energy, the Phoenix Nuclear Power Plant, its ownership consortium, personnel, and license/docket numbers are invented. Reactor designs (Westinghouse 4-loop PWR, US EPR) and named equipment vendors (Westinghouse, Framatome, GE, Arabelle Solutions/EDF, Mitsubishi Heavy Industries, Japan Steel Works, Emerson, Siemens, Rolls-Royce mtu, Fairbanks Morse, Holtec, Combustion Engineering) are real; their attribution here is a plausible fictional configuration, not a statement of any actual contract or installation. Every event happening with any, even far involvement of Desert Horizons Energy, Phoenix Nuclear Power Plant, are fictional.
*NOTE 2: The document was actually written on 04.08.2026. **All** events happening past that date are fictional.*

1. Overview

The Phoenix Nuclear Power Plant is a two-unit nuclear generating station located roughly 55 miles southwest of downtown Phoenix, in the open Sonoran Desert of southwestern Maricopa County, near the unincorporated locality of Ocotillo Flats. It is owned by a seven-member consortium and operated by Desert Horizons Energy Corporation (DHE), an investor-owned utility headquartered in Phoenix.

The station comprises two dissimilar pressurized-water reactors of different generations:

	Unit 1	**Unit 2**
Reactor type	Westinghouse 4-loop PWR	US EPR (Framatome)
Commercial operation	November 1999	April 2027
Licensed thermal power	3,612 MWt	4,590 MWt
Gross electrical output	~1,335 MWe	~1,710 MWe
Net electrical output	~1,270 MWe	~1,600 MWe
NRC docket	50-547	52-046
NRC license	NPF-91	NPF-98

Combined net station capacity is approximately 2,870 MWe, making Phoenix the second-largest nuclear station in the Western Interconnection by nameplate capacity and the largest single source of carbon-free electricity in Desert Horizons Energy's service territory.

Like the nearby Palo Verde station, Phoenix has no adjacent natural body of cooling water and is the only major thermal plant in the region cooled almost entirely by reclaimed municipal wastewater.

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2. Lore & History

2.1 Origins (1973–1985)

Phoenix traces its origins to the energy anxiety of the early 1970s. In 1973, a consortium of Southwestern utilities — then trading under the working name Salt River Valley Nuclear Project — optioned a 6,400-acre parcel of federal and state trust land on the Ocotillo Flats, attracted by its geologic stability, remoteness, low population density, and proximity to the fast-growing Phoenix load center.

The original plan called for three identical Westinghouse 4-loop units (Units 1, 2, and 3). Ground was broken on Unit 1 in 1978. As with most American nuclear projects of the era, the schedule collapsed under the combined weight of post-Three Mile Island regulatory revision, interest-rate escalation, and softening load growth. In 1984, the consortium cancelled Units 2 and 3 outright, writing off roughly $1.1 billion in sunk cost. The half-excavated Unit 2 foundation was backfilled and left fallow — locally nicknamed "the grave."

2.2 The long build and first criticality (1985–1999)

Unit 1 construction limped forward through the late 1980s and early 1990s, repeatedly re-baselined. The reorganized ownership group rebranded the project Phoenix Nuclear Power Plant in 1991 — a deliberate double meaning: the desert city of Phoenix, and the mythological bird rising renewed from its own ashes, a nod to a project that had very nearly died.

Unit 1 achieved:

- Initial fuel load: June 1999
- Initial criticality: August 1999
- Grid synchronization: September 1999
- Commercial operation: November 1999

At the time, Unit 1 was among the last conventional light-water reactors to enter service in the United States before the two-decade construction hiatus. Desert Horizons Energy Corporation was formed in 1996 out of the consortium's operating arm to hold the NRC operating license and run the plant.

2.3 Rebirth as an EPR (2008–2027)

Interest in a second unit revived during the late-2000s "nuclear renaissance," accelerated by Arizona's rising desert-driven summer load, data-center demand, and state decarbonization targets. Rather than resurrect the cancelled Westinghouse design, DHE and its partners pursued a modern Generation III+ reactor.

- 2008: DHE files a Combined License (COL) application with the NRC for a single US EPR on the preserved Unit 2 footprint.
- 2014: NRC issues Combined License NPF-98 (Docket 52-046).
- 2016: First nuclear concrete poured on the rehabilitated Unit 2 basemat — literally atop "the grave" excavated four decades earlier.

- 2019–2024: Extended construction, including the characteristic EPR schedule pressures (large-forging logistics, I&C qualification, and first-of-a-kind-in-the-Americas commissioning).
- December 2026: Initial fuel load.
- February 2027: First criticality.
- April 2027: Commercial operation declared.

Unit 2 is the first EPR to operate in the Western Hemisphere's Americas and gave the "phoenix" branding a second, literal meaning: a reactor rising from a foundation abandoned in 1984.

2.4 Timeline summary

Year	Milestone
1973	Site optioned; three-unit project announced
1978	Unit 1 groundbreaking
1984	Units 2 & 3 cancelled
1991	Project renamed "Phoenix Nuclear Power Plant"
1996	Desert Horizons Energy Corporation formed
1999	Unit 1 commercial operation (November)
2008	Unit 2 (EPR) COL application filed
2014	Unit 2 Combined License issued
2016	Unit 2 first concrete
2017	Unit 1 steam generator replacement + power uprate
2019	Unit 1 license renewal (to 2059)
2027	Unit 2 commercial operation (April)

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3. Ownership & Governance

Phoenix is jointly owned by seven regional entities and operated by DHE under a joint participation agreement. Ownership shares are identical across both units.

Participant	Type	Share
Desert Horizons Energy Corporation (operator)	Investor-owned utility (AZ)	29.1%
Grand Canyon Power & Water District	Public power (AZ)	17.5%
Copperline Electric Company	Investor-owned utility (S. AZ)	15.8%
Pacifica Edison	Investor-owned utility (S. California)	15.0%
Rio Bravo Electric Cooperative	Cooperative (New Mexico)	10.2%
Coastal Municipal Power Alliance	Municipal joint agency (CA)	8.4%
Los Alamitos Water & Power	Municipal utility (CA)	4.0%

Regulatory oversight: U.S. Nuclear Regulatory Commission (NRC) — Region IV, Arlington, Texas. Economic regulation of DHE's retail rates falls to the Arizona Corporation Commission.

Operating organization: DHE Nuclear (a business unit of DHE), led by a Chief Nuclear Officer reporting to the DHE board's Nuclear Committee. The station is run day-to-day by a Site Vice President with separate Plant Managers for each unit sharing common services.

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4. Site Description

- Location: Ocotillo Flats, southwestern Maricopa County, ~55 mi SW of Phoenix; ~28 mi from the nearest incorporated town.
- Total site: ~6,400 acres owned; ~3,300-acre exclusion area; low-population surroundings (federal desert land and state trust rangeland).
- Elevation: ~1,050 ft (320 m).
- Terrain/geology: Flat alluvial basin over stable Basin-and-Range sediments; nearest capable fault studied and screened during licensing. Design basis accommodates regional seismicity; Unit 2 (EPR) is designed to a hardened seismic envelope with aircraft-impact-resistant structures.
- Climate driver: Extreme summer heat (routinely >45 °C/113 °F) is the dominant environmental design factor — it depresses condenser efficiency and drives ultimate heat sink and switchyard equipment ratings.

The absence of a river, lake, or coastline is the defining site constraint and dictates the water strategy in Section 7.

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5. Unit 1 — Westinghouse 4-Loop PWR

5.1 Nuclear steam supply system (NSSS)

- Reactor type: Westinghouse four-loop pressurized water reactor
- NSSS vendor: Westinghouse Electric Corporation
- Licensed core thermal power: 3,612 MWt (original 3,411 MWt; uplifted via measurement-uncertainty recapture + stretch uprate, 2017)

- Reactor coolant loops: 4
- Reactor coolant pumps: 4 × Westinghouse Model 93A (single-stage, vertical, shaft-seal)
- Steam generators: 4 ×
    - Original: Westinghouse Model F (Alloy 600 tubing)
    - Replacement (2017): Framatome replacement steam generators with thermally treated Alloy 690TT tubing
- Pressurizer: Westinghouse; normal RCS pressure ~2,250 psia
- Reactor vessel: Low-alloy steel with stainless cladding; forged and fabricated by Combustion Engineering (Chattanooga)
- Control: 53 rod cluster control assemblies; soluble boron chemical shim

5.2 Core & fuel

- Fuel assemblies: 193, 17×17 lattice
- Fuel: UO_2 pellets, enrichment ≤ 4.95 % U-235
- Fuel vendor: Westinghouse (primary); qualified second source, Framatome, since 2013
- Fuel cycle: 18-month
- Cladding: Advanced zirconium alloy (ZIRLO / Optimized ZIRLO)

5.3 Containment

- Type: Large dry containment — post-tensioned, prestressed reinforced concrete cylinder with hemispherical dome and a welded steel liner
- Design pressure: ~60 psig
- Engineered safeguards: Two-train emergency core cooling (high- and low-pressure injection, accumulators), containment spray, quench-tank / RWST inventory

5.4 Turbine island & electrical

- Turbine-generator: General Electric (GE) tandem-compound, six-flow, 1,800 rpm (four-pole, 60 Hz)

- Uprate note: Low-pressure turbine rotors and last-stage buckets replaced with an efficiency retrofit by Mitsubishi Heavy Industries (MHI) during the 2017 outage.
- Main generator: Hydrogen-inner-cooled, ~1,500 MVA, 24 kV
- Main condenser: Titanium-tubed, single-pressure (selected for reclaimed-water chemistry)
- Main transformers: three single-phase GSU banks stepping 24 kV → 525 kV
- Emergency diesel generators: 2 × Fairbanks Morse Colt-Pielstick PC2.5 (Class 1E, ~5.5 MWe each), plus one shared spare-parts pool with Unit 2

5.5 Instrumentation & control

- As-built (1999): Analog protection with Westinghouse Eagle-21 process protection racks
- Digital upgrades:
    - Reactor protection & engineered safety features actuation migrated to Westinghouse Common Q (Class 1E)
    - Non-safety plant control and turbine control migrated to Emerson Ovation distributed control system
- Main control room: Hybrid analog/digital following upgrades; dedicated remote shutdown panel

5.6 Licensing

- NRC operating license: NPF-91 (Part 50), Docket 50-547
- Original term: 40 years (issued 1999, to 2039)
- License renewal: Approved 2019, extended 20 years to 2059
- Subsequent renewal: Feasibility study underway (potential extension to 2079)

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6. Unit 2 — US EPR (Framatome)

6.1 Nuclear steam supply system

- Reactor type: US EPR — Generation III+ four-loop PWR

- NSSS / reactor vendor: Framatome (formerly AREVA NP)
- Licensed core thermal power: 4,590 MWt
- Reactor coolant loops: 4
- Reactor coolant pumps: 4 × Framatome (vertical single-stage, high-inertia flywheel for extended coastdown)
- Steam generators: 4 × Framatome axial-economizer design, Alloy 690TT tubing
- Pressurizer: Framatome — notably large volume (~75 m^3) for load-transient tolerance
- Reactor pressure vessel: Heavy forgings supplied by Japan Steel Works (JSW); vessel assembled and qualified by Framatome. Designed for a 60-year service life with generous fluence margins.

6.2 Core & fuel

- Fuel assemblies: 241, 17×17 lattice
- Fuel: UO_2, enrichment ≤ 5 % U-235; design also qualified for MOX capability
- Fuel vendor: Framatome
- Fuel cycle: 18–24 month capability
- Neutron reflector: Heavy stainless-steel radial reflector — improves neutron economy and reduces vessel fluence

6.3 Safety architecture — the EPR difference

The EPR's signature is four-train redundancy (N+2) with strong physical separation:

- Four independent safeguard divisions, each in its own building; two of the four divisions housed in bunkered structures separated across the reactor building for protection against internal and external hazards.
- Double-wall containment:
    - Inner shell: prestressed concrete with a steel liner (leak-tight boundary)
    - Outer shell: thick reinforced concrete providing aircraft-impact and external-event protection; annular space maintained at slight negative pressure and filtered

- Core catcher: dedicated spreading compartment beneath the reactor cavity to retain and cool molten core debris in a severe accident — a defining EPR feature.
- Passive autocatalytic recombiners (PARs): hydrogen control without power.
- In-containment refueling water storage tank (IRWST): large internal water inventory feeding safety injection and containment heat removal.
- Dedicated severe-accident heat removal system.

6.4 Turbine island & electrical

- Turbine-generator: Arabelle half-speed (1,800 rpm, 60 Hz) saturated-steam turbine — the largest turbine class in commercial nuclear service.
    - Supply/servicing lineage (realistic): The Arabelle line originated with Alstom, passed to GE Steam Power, and — following EDF's 2024 acquisition of GE Vernova's steam-nuclear activities — new-build supply now sits with Arabelle Solutions (an EDF subsidiary). Because that transaction excluded servicing in the Americas, Unit 2's in-service turbine-island maintenance is contracted to GE Vernova's retained Steam Power services business. Phoenix thus straddles both sides of the split: French-supplied machine, US-serviced.
- Main generator: ~1,800 MVA, hydrogen/water-cooled, 27 kV
- Main condenser: Titanium-tubed, three-shell, matched to reclaimed-water duty
- Main transformers: single-phase GSU bank, 27 kV → 525 kV
- Emergency power:
    - 4 × main emergency diesel generators (one per safeguard division), supplied by Rolls-Royce mtu (~8 MWe each, Class 1E)
    - 2 × station-blackout ("SBO") diesel generators, diverse and independent, for beyond-design-basis loss of AC power

6.5 Instrumentation & control

- Safety I&C (Protection System): Framatome TELEPERM XS (TXS) — the qualified digital safety platform used across the EPR fleet
- Priority & Actuator Control System (PACS): hardwired priority logic between safety and non-safety commands
- Process/plant control (non-safety): distributed control system on the Siemens SPPA-T3000 platform

- Diverse actuation system: independent backup for anticipated-transient-without-scram protection
- Main control room: Fully computerized, screen-based, with a hardwired safety-grade backup panel

6.6 Licensing

- NRC Combined License: NPF-98 (Part 52), Docket 52-046
- COL issued: 2014
- Commercial operation: April 2027
- License term: 40 years, to ~2066 (renewal-eligible)

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7. Cooling & Water Management

Phoenix's defining engineering challenge is cooling a ~2,900 MWe station in a desert with no river or lake.

- Primary cooling-water source: Reclaimed municipal wastewater (Class A+ reclaimed effluent) piped ~35 miles from metro-Phoenix water reclamation facilities under long-term supply agreements.
- On-site polishing: A dedicated Water Reclamation Facility further treats incoming effluent to protect condenser and cooling-tower chemistry against scaling and biofouling.
- Heat rejection: Mechanical-draft cooling towers — one bank per unit — reject waste heat to the atmosphere by evaporation.
- Backup / makeup: On-site groundwater wells (permit-limited) provide emergency makeup only.
- Discharge: Effectively zero liquid discharge — cooling-tower blowdown is routed to lined evaporation ponds; no effluent is released to surface waters.
- Ultimate heat sink (safety): Separate, seismically qualified essential spray-pond / cooling-tower systems dedicated to each unit's safety loads, independent of the circulating-water system.

Reclaimed water reuse makes Phoenix one of the most water-efficient large thermal stations in North America on a per-MWh basis, and a frequent reference case for arid-region power siting.

8. Switchyard & Grid Interconnection

- Interconnection: WECC — Western Interconnection; the plant ties into the heavily traded regional transmission corridor near the Palo Verde/Hassayampa hub area.
- Switchyard voltages: 525 kV (bulk export) and 230 kV (regional / start-up power).
- Configuration: Breaker-and-a-half 525 kV yard for high reliability; multiple independent offsite-power feeds satisfy NRC station-blackout and offsite-power requirements.
- Outgoing lines: 525 kV circuits toward the Phoenix metro load center and westward toward the California interties; 230 kV lines serve regional distribution and provide diverse startup sources.
- Black-start / offsite reliability: Two physically independent offsite power sources per unit, plus the on-site EDGs and (Unit 2) SBO diesels.

9. Spent Fuel & Radioactive Waste

- Spent fuel pools: Each unit has a dedicated, seismically qualified spent-fuel pool with high-density racks and safety-grade cooling.
- Dry storage (ISFSI): A shared on-site Independent Spent Fuel Storage Installation using Holtec HI-STORM vertical dry-cask systems; loaded casks are transferred to the licensed concrete pad as pool capacity is managed.
- Low-level radioactive waste: Processed, volume-reduced, and shipped to a licensed off-site disposal facility.
- High-level waste policy: Consistent with the national spent-fuel situation, no federal repository is available; spent fuel is stored on-site indefinitely under NRC license, with the ISFSI sized for the operating life of both units.

10. Operations & Workforce

- Staffing: ~2,700 permanent DHE Nuclear employees across both units and shared services.

- Outage surge: Refueling outages add up to ~1,800 temporary craft and specialist contractors on site.
- Refueling: Units are refueled on staggered ~18-month cycles so both are rarely offline together; outages are scheduled for spring and fall to avoid peak summer demand.
- Capacity factor: Unit 1 has sustained a lifetime capacity factor above 90%. Unit 2 is in its initial fuel cycle, ramping through first-cycle monitoring and warranty testing as of mid-2027.
- Industry oversight: Member of the Institute of Nuclear Power Operations (INPO); participates in WANO peer reviews.
- Training: On-site licensed-operator training center with full-scope control-room simulators for both the Unit 1 (Westinghouse) and Unit 2 (EPR) designs — the dual-simulator arrangement is unusual and necessary given the two very different control philosophies.

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11. Emergency Preparedness & Community

- Emergency planning zones: 10-mile plume-exposure EPZ and 50-mile ingestion-pathway EPZ, coordinated with Maricopa County and Arizona state emergency management.
- Population advantage: The remote desert siting means the plume EPZ contains a very small resident population — a significant safety and licensing advantage.
- Economic footprint: PNPP is among the largest employers and taxpayers in southwestern Maricopa County; DHE funds regional STEM education, first-responder training, and desert habitat conservation offsets around the site.

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12. Vendor Summary

System	Unit 1 (PWR)	Unit 2 (US EPR)
Reactor / NSSS	Westinghouse	Framatome
Reactor vessel forgings/fab	Combustion Engineering	Japan Steel Works (fab: Framatome)
Steam generators	Westinghouse (orig.) → Framatome (2017)	Framatome

System	Unit 1 (PWR)	Unit 2 (US EPR)
Reactor coolant pumps	Westinghouse Model 93A	Framatome
Fuel	Westinghouse (2nd source Framatome)	Framatome
Turbine-generator	GE (LP retrofit: MHI)	Arabelle (supply: Arabelle Solutions/EDF; Americas service: GE Vernova)
Safety I&C	Westinghouse Common Q	Framatome TELEPERM XS
Control DCS	Emerson Ovation	Siemens SPPA-T3000
Emergency diesels	Fairbanks Morse Colt-Pielstick	Rolls-Royce mtu (+ SBO diesels)
Dry-cask storage	Holtec HI-STORM (shared ISFSI)	Holtec HI-STORM (shared ISFSI)

13. Quick Reference Fact Sheet

- Name: Phoenix Nuclear Power Plant (PNPP)
- Operator: Desert Horizons Energy Corporation
- Location: Ocotillo Flats, SW Maricopa County, Arizona (~55 mi SW of Phoenix)
- Units: 2 (both PWR)
- Unit 1: Westinghouse 4-loop, 3,612 MWt / ~1,270 MWe net, online Nov 1999, license to 2059
- Unit 2: US EPR, 4,590 MWt / ~1,600 MWe net, online Apr 2027, license to ~2066
- Combined net capacity: ~2,870 MWe
- Cooling: Reclaimed municipal wastewater + mechanical-draft cooling towers; zero liquid discharge
- Grid: WECC Western Interconnection; 525 kV / 230 kV switchyard
- Regulator: NRC Region IV
- Notable firsts: First EPR to operate in the Americas; Unit 2 built atop the foundation of a unit cancelled in 1984